

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		Molecules drawn with a peptide bond correct (1) Molecule of <u>water/H₂O</u> also produced (1)	2			2		
		(ii)		peptide bond	1			1		
	(b)	(i)		Quaternary	1			1		
		(ii)		α helix (1) By hydrogen bonding (1) Ignore reference to peptide bonds	2			2		
		(iii)		Two - one for each polypeptide chain (1)			1	1		
	(c)	(i)		360 / 363 / 366 (1)		1		1	1	
		(ii)		Any five (×1) from: 1. DNA/ (pre)mRNA contains both introns and exons (1) 2. The entire gene is transcribed (1) 3. Introns are non-coding/ only exons are coding (1) 4. The (pre)mRNA is then edited to remove the introns / introns are removed from (pre)mRNA/ only leave exons(1) 5. 'pre-edited' (pre) mRNA molecules and 'post-edited' mRNA molecules have different masses/ OWTTE (1) 6. only the 'edited' mRNA is used {in translation / to produce the protein} (1)		2	3	5		
				Question 4 total	6	3	4	13	1	